Page 1: Visual Structural Audits

Guidelines for structural integrity audits across industrial frames, concrete supports, and metallic enclosures. Inspect critical load-bearing joints every 30 days.

Page 2: Structural Crack Mitigation

Defect Type: Structural Crack.

Symptom: Linear fracture propagation on structural casing or welded support seams.

Action Required: Halt structural loads immediately. Perform ultrasonic non-destructive testing (NDT) to assess depth. V-groove the crack and execute a full-penetration weld repair using E7018 electrodes. Reinforce joint with structural backing plate. Priority: Critical.

Page 3: Weld Repair Verification

Procedure: Checking a repaired weld.

After a weld repair let the joint cool to ambient temperature, then run dye penetrant or magnetic particle testing over the full seam. Record the crack length before and after repair. A repair that shows any remaining linear indication must be ground out and repeated. Priority: High.

Page 4: Re-commissioning and Sign-off

Procedure: Returning a structure to service.

A certified inspector must sign the re-commissioning form after NDT passes. Apply load in 25 percent steps and watch the repaired joint for new cracking. Update the inspection schedule to every 7 days for the next 90 days.